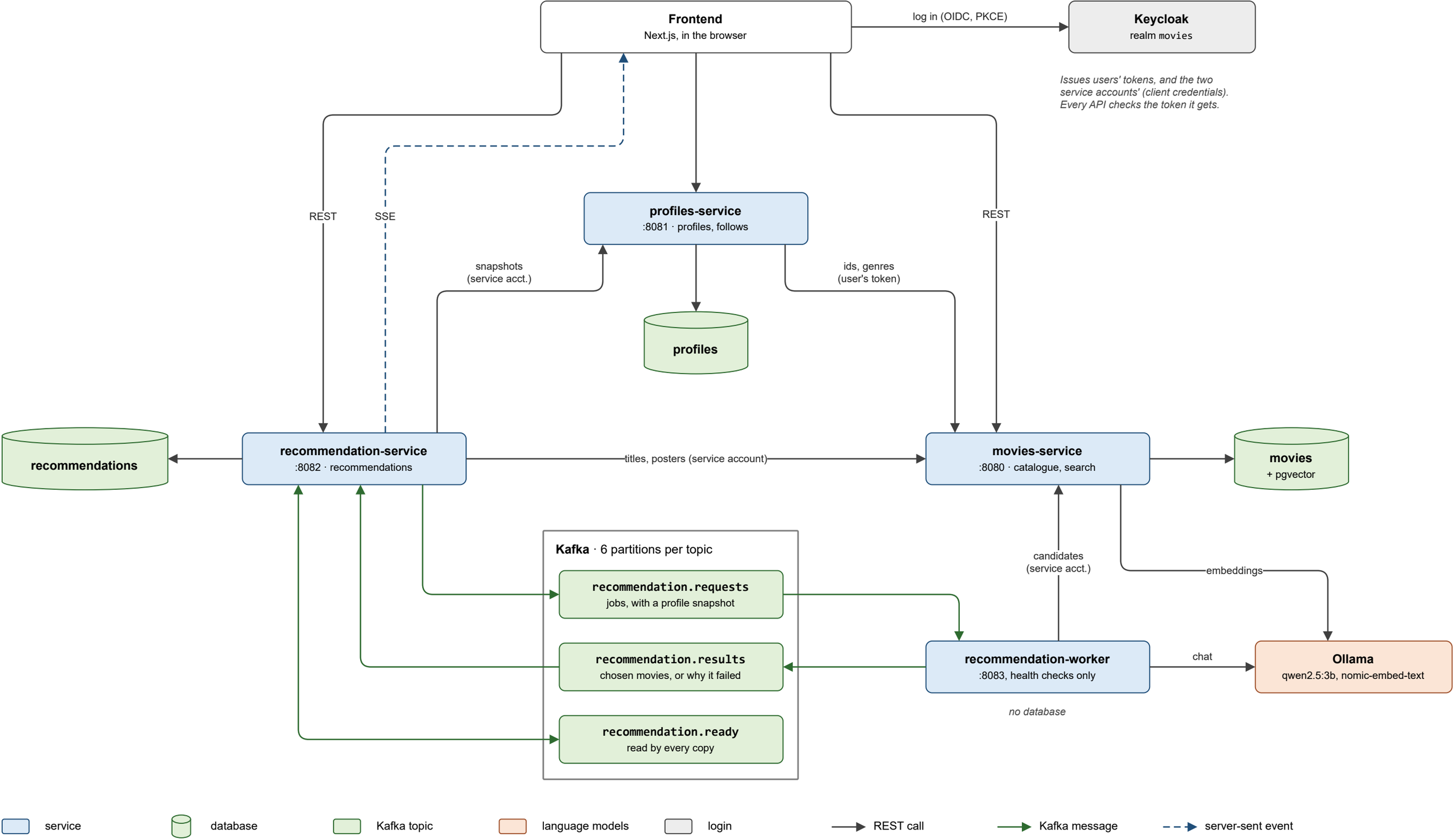

Frontend
Next.js, in the browser
log in (OIDC, PKCE)
Keycloak
realm movies
Issues users' tokens, and the two service accounts' (client credentials). Every API checks the token it gets.
REST
SSE
REST
profiles-service
:8081 · profiles, follows
snapshots (service acct.)
ids, genres (user's token)
profiles
recommendations
recommendation-service
:8082 · recommendations
titles, posters (service account)
movies-service
:8080 · catalogue, search
movies
+ pgvector
Kafka · 6 partitions per topic
recommendation.requests
jobs, with a profile snapshot
recommendation.results
chosen movies, or why it failed
recommendation.ready
read by every copy
candidates (service acct.)
embeddings
recommendation-worker
:8083, health checks only
chat
Ollama
qwen2.5:3b, nomic-embed-text
no database
service
database
Kafka topic
language models
login
REST call
Kafka message
server-sent event